

HOSPITAL AUTOMATION SYSTEM

TEAM MEMBERS

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CODE:

```
1  #include <Servo.h>
2  #include <LiquidCrystal_I2C.h>
3
4  // ----- Automatic Door -----
5  #define doorEchoPin 11
6  #define doorTrigPin 10
7  Servo doorServo;
8  int doorDistance = 0;
9  bool doorOpen = false;
10
11 // ----- Smoke Detection -----
12 #define smokePin A0
13 #define smokeBuzzerPin 4
14 int smokeThreshold = 400;
15 LiquidCrystal_I2C lcd(0x27, 16, 2);
16
17 // ----- Cold Storage -----
18 #define tempPin A1
19 #define tempBuzzerPin 5
20
21 // ----- Hand Sanitizer -----
22 #define sanTrigPin 6
23 #define sanEchoPin 7
24 Servo sanitizerServo;
25
```

```

24 servoSanitizerServo,
25
26 void setup() {
27     Serial.begin(9600);
28
29     // Door Setup
30     pinMode(doorTrigPin, OUTPUT);
31     pinMode(doorEchoPin, INPUT);
32     doorServo.attach(9);
33     doorServo.write(0); // Closed position
34
35     // Smoke Setup
36     pinMode(smokeBuzzerPin, OUTPUT);
37
38     // LCD Setup
39     lcd.init();
40     lcd.backlight();
41
42     // Cold Storage Setup
43     pinMode(tempBuzzerPin, OUTPUT);
44
45     // Sanitizer Setup
46     pinMode(sanTrigPin, OUTPUT);
47     pinMode(sanEchoPin, INPUT);
48     sanitizerServo.attach(3);
49     sanitizerServo.write(0); // Initial position
50 }
51
52 void loop() {
53     handleDoor();
54     handleSmoke();
55     handleTemperature();
56     handleSanitizer();
57     delay(500);
58 }
59
60 // ===== FUNCTIONS =====
61
62 void handleDoor() {
63     digitalWrite(doorTrigPin, LOW);
64     delayMicroseconds(2);
65     digitalWrite(doorTrigPin, HIGH);
66     delayMicroseconds(10);
67     digitalWrite(doorTrigPin, LOW);
68
69     long duration = pulseIn(doorEchoPin, HIGH);
70     doorDistance = duration * 0.034 / 2;
71
72     Serial.print("Door Distance: ");
73     Serial.println(doorDistance);
74

```

```
75     if (doorDistance < 100 && !doorOpen) {
76         doorServo.write(90); // Open
77         doorOpen = true;
78         delay(5000);          // Stay open
79     } else if (doorDistance >= 100 && doorOpen) {
80         doorServo.write(0); // Close
81         doorOpen = false;
82     }
83 }
84
85 void handleSmoke() {
86     int smokeLevel = analogRead(smokePin);
87     Serial.print("Smoke Level: ");
88     Serial.println(smokeLevel);
89
90     if (smokeLevel > smokeThreshold) {
91         lcd.setCursor(0, 0);
92         lcd.print("SMOKE DETECTED! ");
93         digitalWrite(smokeBuzzerPin, HIGH);
94     } else {
95         lcd.setCursor(0, 0);
96         lcd.print("AIR IS CLEAN    ");
97         digitalWrite(smokeBuzzerPin, LOW);
98     }
99 }
```

```
100
101 void handleTemperature() {
102     int reading = analogRead(tempPin);
103     float voltage = reading * (5.0 / 1023.0);
104     float temperature = voltage * 100;
105
106     Serial.print("Temp: ");
107     Serial.println(temperature);
108
109     lcd.setCursor(0, 1);
110     lcd.print("Temp: ");
111     lcd.print(temperature, 1); // 1 decimal place
112     lcd.print("C ");
113
114     if (temperature > 10.0) {
115         digitalWrite(tempBuzzerPin, HIGH);
116     } else {
117         digitalWrite(tempBuzzerPin, LOW);
118     }
119 }
120
121 void handleSanitizer() {
122     digitalWrite(sanTrigPin, LOW);
123     delayMicroseconds(2);
124     digitalWrite(sanTrigPin, HIGH);
```

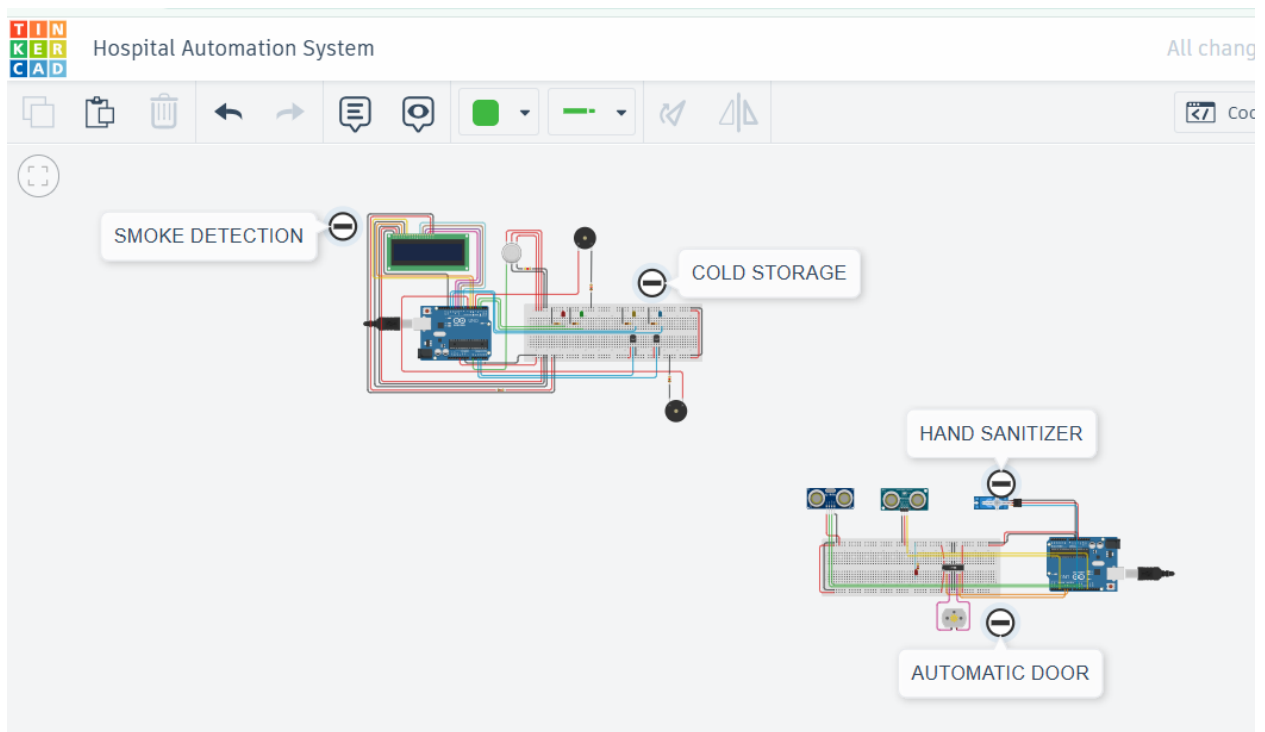
```

125     delayMicroseconds(10);
126     digitalWrite(sanTrigPin, LOW);
127
128     long duration = pulseIn(sanEchoPin, HIGH);
129     int distance = duration * 0.034 / 2;
130
131     Serial.print("Sanitizer Distance: ");
132     Serial.println(distance);
133
134     if (distance > 3 && distance < 10) {
135         sanitizerServo.write(90); // Dispense sanitizer
136         delay(1000);              // Wait for dispensing
137         sanitizerServo.write(0);  // Reset to original position
138         delay(2000);              // Cooldown before next dispense
139     }
140 }

```

DIAGRAM:

(overall diagram)



(overall diagram + code)

TIMEXCAD Hospital Automation System

All changes saved

Code Start Simulation Send To

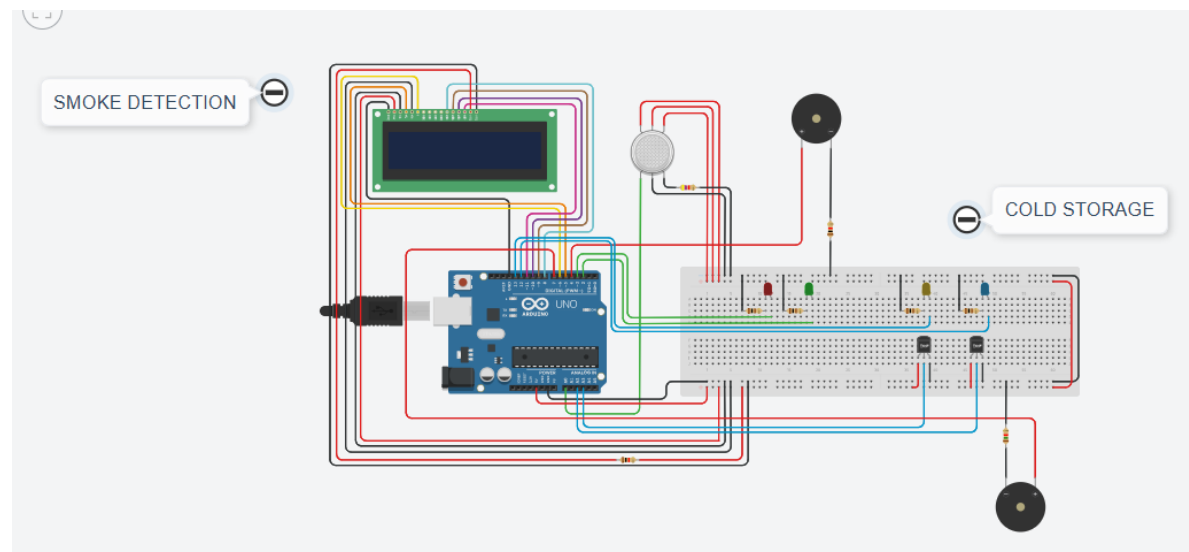
Text 2 (Arduino Uno R3)

```
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21 // ----- Hand Sanitizer -----
22 #define sanTrigPin 6
23 #define sanEchoPin 7
24 Servo sanitizerServo;
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26 void setup() {
27   Serial.begin(9600);
28
29   // Door Setup
30   pinMode(doorTrigPin, OUTPUT);
31   pinMode(doorEchoPin, INPUT);
32   doorServo.attach(9);
33   doorServo.write(0); // Closed position
34 }
```

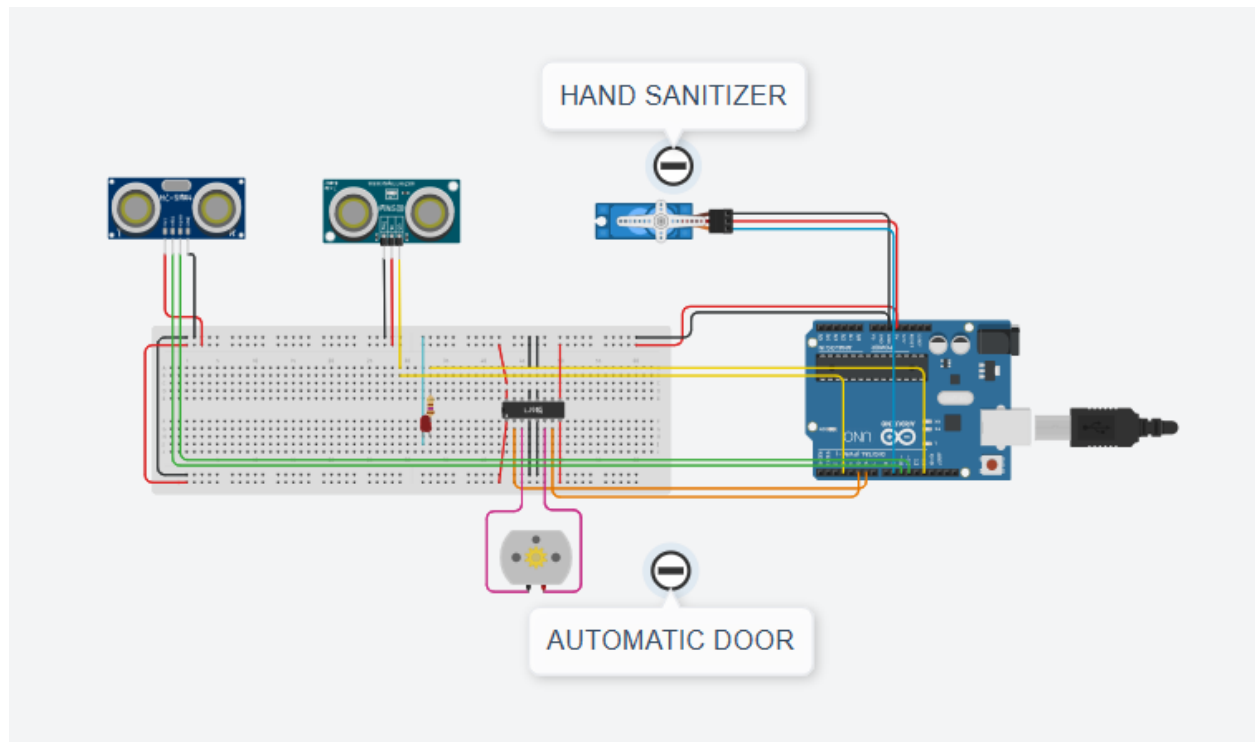
Serial Monitor

The diagram shows an Arduino Uno R3 connected to a breadboard. The breadboard contains a servo motor, a buzzer, and an LCD display. Wires connect the Arduino pins to the components. Labels indicate the functions: SMOKE DETECTION, COLD STORAGE, HAND SANITIZER, and AUTOMATIC DOOR.

1 portion:



2nd portion



.....END.....